

Milestone 1: Project Foundation

June 11, 2026

CS 4632 Course Project: Discrete-Event Simulation of an Electronics Repair Department

by

Carlos Guerrero

Abstract: This project will be a discrete-event simulation of the electronics repair support department where I work. The department repairs printed circuit boards and HVAC/refrigeration controllers. Jobs arrive from advance exchanges, production-line failures, reships, and customer Return Material Authorizations (RMAs). Customer RMAs wait on a physical rack, while the other work types are brought directly to the technicians. The model will represent two technicians, two test stations, priority rules, and interruptions for urgent production work. I will use historical data to compare turnaround time, RMAs, utilization, throughput, and repair outcomes.

Project repository:

<https://github.com/CarlosG100/CS4632-Electronics-Repair>

1 Project Overview

1.1 Domain and Problem

The project is based on an electronics manufacturing support department. The department receives PCBs and complete controller units, analyzes the failures, repairs what it can, and records a final outcome. The workload is not constant because jobs/units arrive at different times and repairs do not all take a uniform amount of time.

The main problem is deciding how priority rules and limited resources affect the different job types. Urgent production failures need a fast response, but interrupting another repair may increase the delay for normal customer work. A simulation is useful because the system changes over time and includes random arrivals and repair times.

The project will answer these questions:

1. How do first-in, first-out (FIFO) and priority with or without preemption affect turnaround time?
2. How do technician specialization and station compatibility affect waiting and capacity?
3. What practical change improves urgent response without creating a large RMA backlog?

1.2 Project Scope

The model will begin when work reaches the department. It will include placement of customer RMAs on the rack, direct handoff of other work, work selection, waiting, service, interruption, completion, and outcome. It will also include the delayed return of a defective advance-exchange unit.

The first model will represent general and specialized technician/station capabilities, but it will not model individual technician efficiency, equipment downtime, parts delays, or rework. About 80 percent of jobs can use either station, while about 20 percent require a compatible technician and specialized station equipment. The model will use working hours rather than calendar hours, so nights and weekends will not count toward repair time.

2 System Description

2.1 Entities, Events, State, and Resources

The entities are the jobs or requests that flow through the department. Each one follows a process based on its source. Customer RMAs are placed on a physical rack and selected in FIFO order. Advance-exchange requests are brought to the department's attention by the RMA administrator. Reships are brought by the receiving department, and production-line failures are brought directly by production personnel. Job attributes include source, priority, failure category, required capability, arrival time, service time, and final outcome.

The main events are:

- a customer RMA is placed on the rack or another work type is handed off;
- a technician begins or resumes work;
- an advance-exchange replacement is picked from stock, programmed, function tested, and handed to shipping;
- a repair is interrupted;
- a repair is completed;
- a job is repaired, scrapped, or returned to vendor (RTV); and
- a defective advance-exchange unit returns later as a customer RMA.

The state variables are the number of RMAs on the rack, direct requests waiting, work-in-process, technician status, station status, resource capability, and the number of completed jobs. The technicians and test stations are passive resources. The model includes two technicians and two stations. Most work can use either technician/station pair, but specialized work must wait for compatible resources.

2.2 Work Selection and Service Process

The proposed priority order is:

1. advance exchange;
2. urgent production-line failure;

3. reship; and
4. customer RMA.

Customer RMAs remain on the rack until a technician is ready for another job, and the technician selects the oldest RMA first. If that RMA requires specialized resources, it waits for the compatible technician and station. Advance exchanges, reships, and production-line failures are direct requests rather than jobs selected from the RMA rack. These direct requests will use the priority order above when more than one needs attention. Only an urgent production failure can interrupt work already in progress. The interrupted repair will keep its remaining repair time and return to its prior waiting position. This allows the model to measure the faster response for urgent work and the extra delay placed on normal work.

An advance-exchange request is preparation work rather than a repair. A technician will pick a replacement unit from stock, take it to a compatible station, program it, and perform a quick function test. The technician will then gather the required paperwork and return the replacement to the shipping department. The station is only occupied during programming and testing. When the defective field unit returns later, it will be created as a separate normal-priority customer RMA repair job.

A reship is brought to the department by receiving. A technician will inspect the unit, perform a function check at a compatible station, gather the new shipment paperwork, and return it to shipping. Production-line personnel bring failed units directly to the department for analysis and repair.

2.3 Core Models and Concepts

The project will use three main ideas:

1. **Discrete-event simulation.** The simulation will use next-event time advance, meaning the clock jumps to the time of the next scheduled event, such as an arrival, repair start, interruption, or completion [1].
2. **FIFO work selection and priority rules.** Customer RMAs will be selected from the rack in FIFO order. Resource compatibility will be checked after selection. Direct requests will be represented separately, with production failures allowed to interrupt lower-priority work. This adapts Cobham's priority waiting-line concepts to the department's actual work-selection process

[3].

3. **Random input models.** Historical data will be used for arrival times, repair times, advance-exchange return delays, and repair outcomes [4].

2.4 Assumptions

- Approximately 80 percent of jobs can use either station and either technician. The remaining 20 percent require a compatible specialization.
- Customer RMAs are selected from the physical rack in FIFO order.
- Advance exchanges, reships, and production failures are brought directly to the technicians and are not selected from the RMA rack.
- Each active repair uses one technician and one compatible station. An advance-exchange request uses a technician throughout preparation, but it only uses a compatible station during programming and the quick function test.
- Interrupted repairs will continue from their remaining repair time.
- Replacement stock will be assumed available, and the selected replacement will pass its quick function test in the first version.
- The simulation clock and turnaround measurements will use department working hours.
- Parts inventory, rework, equipment downtime, and individual technician efficiency differences are outside the first version.

3 Implementation Approach

3.1 Language and Tools

This will be a discrete-event simulation. I will write it in Python 3 using SimPy because Python is readable and SimPy supports event timing, resources, and interruptions. I will use Visual Studio Code for development. I will write the job flow, FIFO rack-selection rules, direct-request priority logic, compatible resource use, interruption logic, delayed RMA return, metrics, and experiments.

3.2 Historical Data

The historical fields are job number, source, priority, arrival time, repair start and end time, general failure category, required capability, outcome, and advance-exchange request, preparation start, programming/test, shipping-handoff, and return times.

For this model, repair service time means active working time between repair start and completion. Time outside department working hours and known waits for parts will be removed.

3.3 Outputs and Experiments

Recorded measures will include RMAs, waiting direct requests, work-in-process, turnaround time, waiting time, advance-exchange preparation time, resource use, throughput, interruptions, and repair outcomes.

Table 1: Planned simulation scenarios

ID	Scenario	Purpose
0	Two technicians and two compatible stations with preemption	Baseline system
1	Priority without preemption	Measure the effect of interruption
2	Handle all waiting work FIFO	Measure the effect of direct-request priority rules
3.1	Add one technician	Test technician capacity and skill coverage
3.2	Add one station	Test station capacity and equipment coverage
4	Three technicians and three stations	Test a balanced capacity increase
5	Higher arrival volume	Test a workload surge
6	Add interruption overhead	Test the cost of preemption

Each scenario will be run several times. I will compare average results and confidence intervals instead of using only one simulation run.

3.4 Eight-Week Timeline

Week	Dates	Planned Work
1	Jun. 3–7	Select project and review initial course work.
2	Jun. 8–14	Complete M1: proposal, literature review, repository, and UML diagrams.
3	Jun. 15–21	Prepare historical inputs and build the basic ob flow.
4	Jun. 22–28	Complete M2: initial implementation with priorities and technician/station resources.
5	Jun. 29–Jul. 5	Add compatible resources, advance exchanges, reships, and process details.
6	Jul. 6–12	Complete M3: full implementation with conditional waiting and interruptions.
7	Jul. 13–19	Complete M4: run experiments, analyze outputs, and validate the model.
8	Jul. 20–23	Complete M5: final report, results, repository, and video.

4 Literature Review

- Banks et al. explain discrete-event simulation through entities, queues, resources, events, and next-event time advance [1]. They also identify electronics assembly operations as a manufacturing use of simulation. This matches my project because controller and PCB jobs wait for limited resources, receive service, and leave with an outcome. I will adapt the basic service-system structure by adding an RMA rack, three direct-request sources, compatible technician/station resources, working hours, and interruptions.
- Law explains process-interaction simulation as following an entity through each step of a system [4]. This fits my project because each job moves through arrival, waiting, service, and completion. I will also use Law's methods to create random inputs from historical data, run each scenario multiple times, compare the results, and check event logs to verify that the simulation works correctly.
- Cobham shows that priority service reduces the waiting time of high-priority work at the cost of increasing the wait for lower-priority work [3]. This is my main tradeoff: an urgent production failure may be handled faster while a normal customer RMA waits longer. I will adapt the priority model by using

FIFO for the RMA rack, direct-request priorities, two compatible service resources, and both preemptive and non-preemptive scenarios. I will also test interruption overhead rather than assuming that stopping and restarting a repair has no cost.

- Little proves the relationship $L = \lambda W$ between average items in a system, throughput rate, and average time in the system [5]. An important part of the result is that it does not depend on one specific arrival distribution, service distribution, number of channels, or queue discipline. I will use this relationship as a reasonableness check by comparing average work-in-process with throughput and average turnaround time from the simulation.
- Borshchev explains that service systems may require particular workers and shared equipment [2]. This supports modeling the 20 percent of jobs that need a specialized technician and station. He also warns that adding every real-world detail can make a model too large to manage. I will represent general and specialized capabilities while leaving individual efficiency, parts inventory, equipment downtime, and rework outside the first version.

5 UML Diagrams

Figure 1 shows the main classes planned for the Python program.

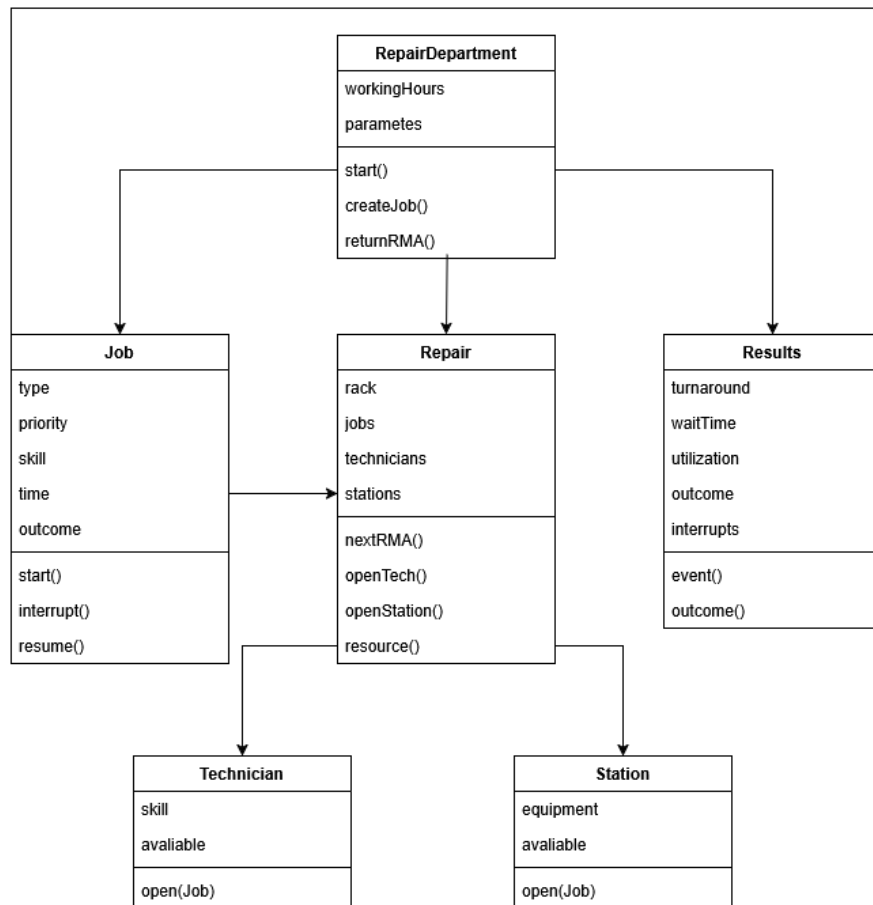


Figure 1: Basic class diagram for the simulation.

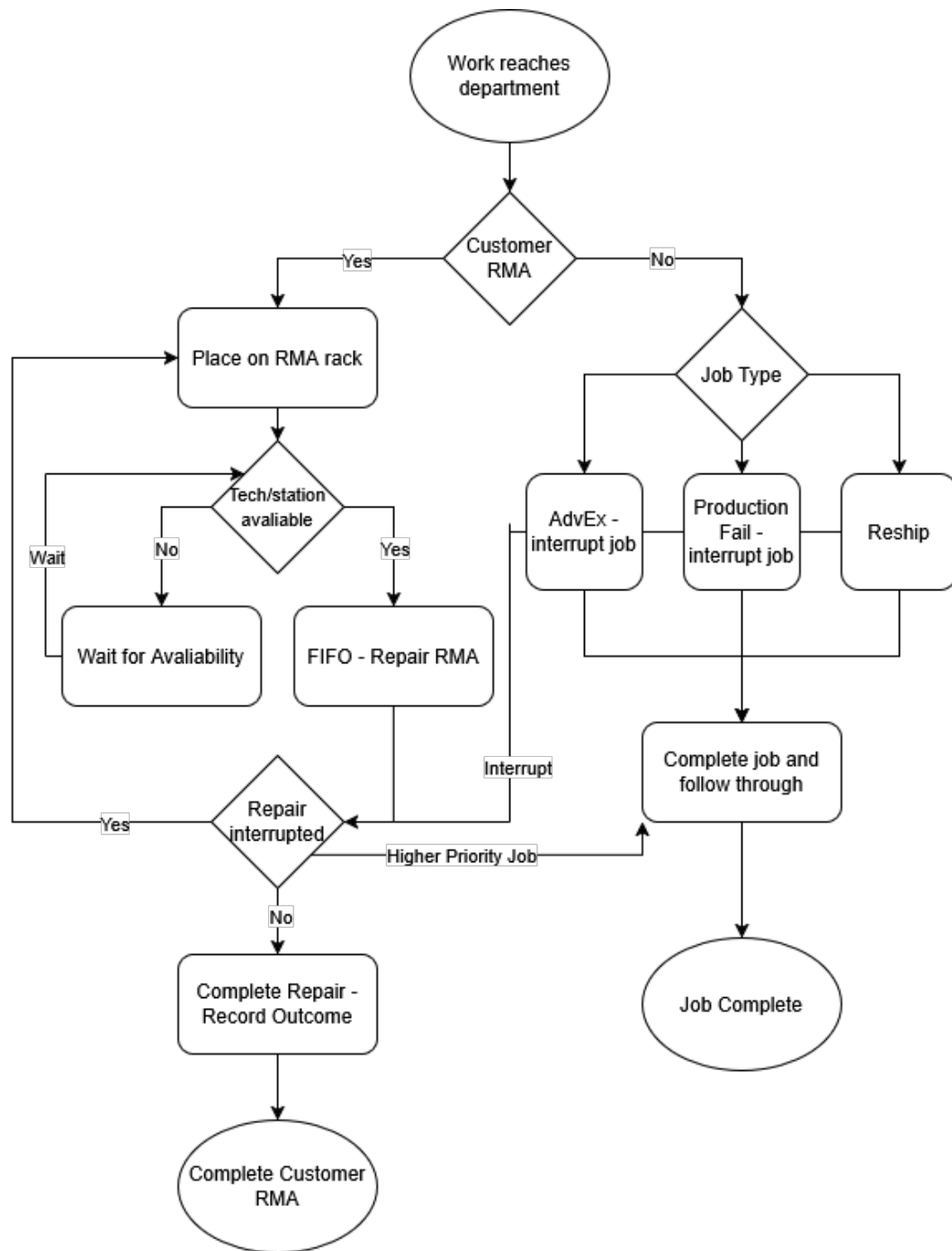


Figure 2: Activity diagram for customer RMAs and directly delivered work.

6 Verification, Validation, and Conclusion

I will verify the model with small test cases, event traces, RMA rack-order checks, and checks that technicians and stations are released correctly. The trace will follow state variables, resource status, and job events after each important event. I will validate the model by comparing simulated arrivals, turnaround times, throughput, and outcomes with the historical data.

The expected result is a model that shows whether the current priority process helps urgent work without causing too much delay for customer RMAs. It should also show whether adding a technician, a station, or a compatible resource combination would provide the most useful improvement.

References

- [1] J. Banks, J. S. C. II, B. L. Nelson, and D. M. Nicol. *Discrete-Event System Simulation*. Prentice Hall, 4 edition, 2005.
- [2] A. Borshchev. Modeling and simulation modeling. In *The Big Book of Simulation Modeling: Multimethod Modeling with AnyLogic 8*, chapter 1. The AnyLogic Company, 2020.
- [3] A. Cobham. Priority assignment in waiting line problems. *Journal of the Operations Research Society of America*, 2(1):70–76, 1954.
- [4] A. M. Law. *Simulation Modeling and Analysis*. McGraw-Hill Education, 5 edition, 2015.
- [5] J. D. C. Little. A proof for the queuing formula: $L = \lambda W$. *Operations Research*, 9(3):383–387, 1961.